

Coincidence and Identity

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1. Do distinct objects coincide?

Possible meanings for 'coincide':

- Spatial: x and y are both located somewhere, and x is located wherever y is, and vice versa.
- Mereological 1: x and y both have at least one part, and an object shares a part with x iff it shares a part with y .

Useful terminology from Fine:

"Extreme monism": whenever x coincides with y , $x=y$.

"Moderate monism": whenever x always coincides with y , $x=y$.

"Mild monism": whenever x necessarily coincides with y , $x=y$.

We can similarly define mild, moderate, and extreme *pluralism* as the negations of extreme, moderate, and mild monism respectively.

2. Two strategies for arguing for pluralism.

First strategy: use "singular terms" (names, demonstratives, definite descriptions).

Here are two representative instances.

An argument for mild pluralism, using definite descriptions and a temporal predicate.

- (1) The statue on my desk coincides with the clay of which it is made.
- (2) The clay was on a riverbed a decade ago.
- (3) The statue was not on a riverbed a decade ago.
- (4) The statue is distinct from the clay (from 2 and 3).
- (5) There are two distinct things that coincide (from 1 and 4).

An argument for moderate pluralism, using names and a modal predicate.

- (1) Goliath permanently coincides with Lump1.
- (2) Lump1 could have been spherical.
- (3) Goliath could not have been spherical.
- (4) Goliath is distinct from Lump1 (from 2 and 3).
- (5) There are two distinct things that permanently coincide (from 1 and 4).

Second strategy: avoid the need to bring in singular terms by using quantifiers.

A quantified argument for mild pluralism, using a modal predicate.

- (1) Every clay statue coincides with a lump of clay.
- (2) Every lump of clay could have been spherical.
- (3) Some clay statue couldn't have been spherical.
- (4) Some clay statue isn't a lump of clay (from 2 and 3).
- (5) There are two distinct things that coincide (from 1 and 4).

Note that it's only the step from (1) and (4) to (5) that invokes distinctive logical rules about *identity*. Some reject the step from (2) and (3) to (4): this requires a revisionary logic of *quantifiers*, not (just) a revisionary logic of identity. In modern logic, it can be formalised like this:

$$\begin{aligned} & \forall x (Fx \rightarrow Gx) \\ \exists x (Hx \wedge \neg Gx) \cdot \\ & \exists x (Hx \wedge \neg Fx) \end{aligned}$$

In Aristotelian logic, it's syllogism of the second figure, in the valid form Baroco.

- Or is that right? One might think that 'is' in 'is an F' means 'is identical to', so that (4) mean 'Some clay statue isn't identical to a lump of clay', analogous to 'Some cat doesn't like a dog'. If so, the above formalization of (4) isn't right, and it should instead be $\exists x (Hx \wedge \neg \exists y (Fy \wedge y = x))$ —which might not be equivalent according to someone with an unorthodox view of identity.

3. Encouragement for monists: some bad arguments

Paradigm 1: gradable adjectives.

- (1) Bruce Wayne is strong.
- (2) Batman is not strong.
- (3) ??? Bruce Wayne is not identical to Batman.

or with quantifiers:

- (1) Some billionaire who lives in this mansion is strong.
- (2) No superhero who lives in this mansion is strong.
- (3) ??? No billionaire who lives in this mansion is a superhero.

Paradigm 2: attitude reports

- (1) Hesperus is believed to be visible in the evening.
- (2) Phosphorus is not believed to be visible in the evening.
- (3) ??? Hesperus is not identical to Phosphorus.

or with quantifiers:

- (1) Every planet visible in the evening is believed to be visible in the evening.
- (2) Some planet visible in the morning is not believed to be visible in the evening.
- (3) ??? Some planet visible in the morning is not visible in the evening.

4. First resistance strategy: deny a premise

NB: you can think that every clay statue is a lump of clay without thinking that every lump of clay that is sometimes a statue is always, or necessarily, a statue. Compare: people and philosophers.

Thomson: premises about part-replacement like (2') and (3') harder to deny than premises about possible or sometime nonexistence.

(2') The statue will have a different hand-shaped part tomorrow.

(3') The lump of clay will not have a different hand-shaped part tomorrow.

What about sentences like 'This lump of clay on the riverbed will later be a statue?' Do they show that (3) is false?

Well we seem to talk both ways. Maybe there is some ambiguity.

- Wiggins: 'is' sometimes means 'constitutes'.
 - But what about 'This statue used to be on a riverbed', which we'll assert when we're in the mood to say things like 'this lump will later be a statue'? 'This statue used to constitute on a riverbed' isn't even grammatical.
- A possible alternative: 'statue' can mean statue₁— something that makes (3) true— but can also mean something like *thing that constitutes a statue*₁.

5. Second strategy for resistance: pleading equivocation

Valid arguments are supposed to "preserve truth". But even when a sequence of sentences fits a valid argument-form, a sequence of utterances of those sentences can take one from truth to falsity, if there is a relevant shift in the interpretation of some expression that's capable of being interpreted in different ways.

This seems a plausible diagnosis of what's going wrong with the superhero arguments. The arguments (understood as sequences of *sentences*) are valid. But 'strong' can be interpreted in different ways; (1) is only true if it's interpreted as something like *strong for*

a billionaire, while (2) is only true if it's interpreted as something like *strong for a superhero*. There's no uniform interpretation where both premises are true.¹

One can also add a story about why the different premises *suggest*, or make natural, different interpretations of the relevant expression.

However, there is strong pressure (if one is going for this strategy) to say that it's linguistically *possible*, even if not so natural, to interpret the premises "uniformly".

- One reason for this: if 'valid' *means* something like 'preserves truth under all linguistically possible uniform interpretations', then an argument whose premises didn't *have* any linguistically possible uniform interpretations would be valid no matter what we plugged in for the conclusion!

How plausible is this as a way of resisting the pro-pluralism arguments?

- On the positive side, it looks like modal words like 'could' admit many interpretations: consider 'I couldn't have attended both meetings since they were held miles apart and there was only a five minute gap between them' versus 'I could have attended both meetings but only if one of them had been postponed / but only if I had a helicopter / ...'.
- On the other hand, such ambiguity is much less independently well motivated in the case of tense operators, or expressions like 'on the riverbed at *t*' where *t* names a time, so the arguments for *mild* pluralism are harder to resist.
- Many philosophers think that they have zeroed in on a specific sense of 'could' etc., the *metaphysical* interpretation, such that equivocation in 'could' can be eliminated by insisting on that interpretation all the way through. But even if they are right, if metaphysical possibility is something we don't talk much about in ordinary life, the motivation for premises like 'Goliath couldn't have been spherical' is not so firm when the modality is metaphysical.

But it's *much* harder to play the equivocation card when we are dealing with arguments where *two* names or free variables occur in the scope of a modal world. For example:

- (1) Lumpl could have outlasted Goliath.
- (2) Goliath could not have outlasted Goliath.
- (3) So, Lumpl is distinct from Goliath.

or (if we prefer to keep identity out of it):

¹ My paper 'Transparency and the Context-sensitivity of Attitude Reports' defends a parallel account of the attitude-report arguments.

- (1) Some lump of clay L on the desk is such that for every statue G on the desk, L could have outlasted G.
- (2) Every statue S on the desk is not such that for every statue G on the desk, S could have outlasted G.
- (3) So, some lump of clay on the desk is not a statue on the desk.

In each case, premise (2) would seem to be true on *every* interpretation of ‘could’, since ‘Everything is such that it could not outlast itself’ is true on every interpretation of ‘could’. (This might be even more forceful if we replace ‘*x* outlasts *y*’ with something like ‘there are times at which *x* exists and *y* does not exist’, where the irreflexivity can be proved in elementary logic.) If so, then so long as there is any sense of ‘could’ that makes premise (1) true, we can just run the argument using that sense throughout and the conclusion will follow.

- A difficult response: there is no *single* reading on which (1) is true. That sentence is like ‘This care is mine, but it is not *mine*’, or ‘Everyone is asleep and being monitored by at least one research assistant’, or like ‘Lumpl could have been spherical but Goliath could not’ (as conceived of by the equivocation-lover). We can use it to convey a truth, but only by speaking non-uniformly.
- Challenge for this: which *words* are we resolving nonuniformly when we use (1) to express a truth?

6. Third strategy for resistance: deny the validity of the arguments.

Here we can distinguish a more radical and a less radical strategy. The more radical strategy already rejects the validity of the argument from (2) and (3) to (4) in the quantified argument (which *prima facie* has nothing to do with identity). The less radical strategy accepts this step but rejects the argument from (1) and (4) to (5) in this argument.

Here’s a toy model for the more radical strategy: suppose we posited that rather than being allowed to mean ‘is strong for a superhero’, ‘is strong for a billionaire’, etc. on any of its occurrences, ‘is strong’ in the context ‘[Quantifier-word] [noun] [modifier] is strong’ is *required* to be interpreted as interchangeable with ‘is strong for a [noun]’.

In trying to think in a disciplined way about what it would be like to reject the validity of the arguments, it’s helpful to restate them in a formal language. Here is a formal version of the Goliath/Lumpl one.

- (1) Coincide(Lumpl, Goliath)
- (2) $\Diamond$ Spherical(Goliath)
- (3) $\neg \Diamond$ Spherical(Lumpl)

- (4) $\neg(\Diamond\text{Spherical}(\text{Lumpl}) \rightarrow \Diamond\text{Spherical}(\text{Goliath}))$ (2, 3, PC)
- (5) $\text{Lumpl} = \text{Goliath} \rightarrow (\Diamond\text{Spherical}(\text{Lumpl}) \rightarrow \Diamond\text{Spherical}(\text{Goliath}))$ (instance of LL)
- (6) $\neg(\text{Lumpl} = \text{Goliath})$ (4, 5, PC)
- (7) $\text{Coincide}(\text{Lumpl}, \text{Goliath}) \wedge \neg(\text{Lumpl} = \text{Goliath})$ (1, 6, PC)
- (8) $\exists y(\text{Coincide}(\text{Lumpl}, y) \wedge \neg(\text{Lumpl} = y))$ (7, EG)
- (9) $\exists x \exists y(\text{Coincide}(x, y) \wedge \neg(x = y))$ (8, EG)

Although one could imagine someone wanting to dispute the propositional logic or EG steps, the central target here is line 5, and the general LL schema $a = b \rightarrow (\Phi[a/z] \rightarrow \Phi[b/z])$ that it instantiates. In the next section we'll consider some other things in the vicinity which those who reject that schema could hold on to (and to which they might prefer to reserve the label 'Leibniz's Law').

What about the quantified argument? Here's a formalisation of that for what it's worth:

- (1) $\forall x(\text{Statue}(x) \rightarrow \exists y(\text{Lump}(y) \wedge \text{Coincide}(y, x)))$
- (2) $\forall y(\text{Lump}(y) \rightarrow \Diamond\text{Spherical}(y))$
- (3) $\exists x(\text{Statue}(x) \wedge \neg\Diamond\text{Spherical}(x))$
- (4) $\forall x(\text{Statue}(x) \rightarrow \exists y(\text{Coincide}(y, x) \wedge \Diamond\text{Spherical}(y)))$ (1, 2)
- (5) $\exists x \exists y(\text{Coincide}(y, x) \wedge \Diamond\text{Spherical}(y) \wedge \neg\Diamond\text{Spherical}(x))$ (3, 4)
- (6) $\forall x \forall y(y = x \rightarrow (\Diamond\text{Spherical}(y) \rightarrow \Diamond\text{Spherical}(x)))$ (Quantified LL)
- (7) $\exists x \exists y(\text{Coincide}(y, x) \wedge \Diamond\text{Spherical}(y) \wedge \neg\Diamond\text{Spherical}(x) \wedge y = x \rightarrow (\Diamond\text{Spherical}(y) \rightarrow \Diamond\text{Spherical}(x)))$ (5, 6)
- (8) $\exists x \exists y(\text{Coincide}(y, x) \wedge \neg(y = x))$ (7)

7. What is Leibniz's Law?

In trying to get to grips with views that reject the steps in the arguments that are specifically to identity, it may be helpful to engage in some mild formalisation and consider some different formal schemas that might be called Leibniz's Law.

	Nominal	Quantified
Substitutional	$a=b \rightarrow (\Phi[a/z] \rightarrow \Phi[b/z])$	$\forall x \forall y(x=y \rightarrow (\Phi[x/z] \rightarrow \Phi[y/z]))$
Predicational	$a=b \rightarrow (Fa \rightarrow Fb)$	$\forall x \forall y(x=y \rightarrow (Fx \rightarrow Fy))$
Higher-order quantified	$\forall Z(a=b \rightarrow (Za \rightarrow Zb))$	$\forall Z \forall x \forall y(x=y \rightarrow (Zx \rightarrow Zy))$

Focus for now on the first two rows.

It's widely thought that the behaviour of names in attitude context gives rise to counterexamples to the Nominal-Substitutional version. For example: Hesperus=Phosphorus and the Babylonians believed Hesperus was visible in the evening, but it is not the case that the Babylonians believed Phosphorus was visible in the evening.

- That view stated in English is compatible with the view that we can construct a formal language in which all versions of LL hold, such that it's possible and perhaps helpful to translate English into that formal language. The translation might involve, for example, transformations somewhat analogous to the one from 'Giorgione was so-called because of his size' to 'Giorgione was called $G^i o^r g^i o^n e$ because of his size'.

Quine, in 'Reference and Modality' uses examples using *definite descriptions* to suggest that Nominal-Substitutional LL is wrong (even for a language without attitude verbs but with modal operators).

But this is a dubious move. Suppose someone argued against the transitivity of identity on the grounds that the premises of the following argument are true but the conclusion is false.

Some philosopher is identical to some American.

Some American is identical to some non-philosopher.

So, some philosopher is identical to some non-philosopher.

This does refute a "Transitivity of Identity" schema stated in English where the rules of the schema allow arbitrary DPs (the grammatical category that includes 'some philosopher', 'the philosopher', and 'Kripke') to be plugged in on either side of 'is identical to'. But the whole virtue of post-Fregean logic is that it gives us a way of

The Predicational version is very weak unless the language has—like English—the resources to construct *complex* predicates like 'was believed by the Babylonians to be visible in the evening'. Many of those who accept the Hesperus/Phosphorus counterexample to the Nominal-Substitutional version still accept the Nominal-Predicational version, including instances like 'Hesperus = Phosphorus $\rightarrow$ (Hesperus was believed by the Babylonians to be visible in the evening $\rightarrow$ Phosphorus was believed by the Babylonians to be visible in the evening)'. They reject the equivalence of ' a is believed by b to be F ' and ' b believes that a is F '.

In formal languages the usual way of forming complex predicates is by *lambda abstraction*. When Φ is a formula and $v_1 \dots v_n$ are variables, $(\lambda v_1 \dots v_n. \Phi)$ is an n -place predicate. (In the interesting cases, $v_1 \dots v_n$ are free in Φ ; some formal languages make this a re-

quirement.) The analogue in such a language for equivalences like ‘ a is believed by b to be F iff b believes that a is F ’ is

Extensional Beta: $(\lambda v_1 \dots v_n. \Phi)(a_1, \dots, a_n) \leftrightarrow \Phi[a_1 / v_1, \dots, a_n / v_n]$

So